

Name

Class

Date

Introduction to Science Practices Test

Total questions: 21

Worksheet time: 22mins

1. Match the following steps of the scientific method to the correct description.

Make an observation ⦿

⦿ Using your 5 sense.

Hypothesis ⦿

⦿ Test your hypothesis.

Experiment ⦿

⦿ Predict what you think the answer to your problem/question will be.

Conclusion ⦿

⦿ Compare results to your hypothesis.

Collect and Analyze Results ⦿

⦿ Organize your data in a table or graph

2. Information you obtain through your senses, such as touch, sound and sight.

a) Question

b) Observation

c) Conclusion

3. Bob read about growing plants in water. He wanted to know how plants could grow without soil.

a) Question

b) Hypothesis

c) Observation

4. Which is true about hypothesis?

a) It summarizes the result on the experiment.

b) It is an educated guess or possible answer to the question.

c) It is the final answer to the question.

5. Testing a hypothesis often involves a(n):

a) Answer

b) Experiment

c) Problem

6. What is the correct hypothesis for "How does exercising affect your health?"

- a) If you exercise, then you will have better health.
- b) If you have better health, then you will exercise.
- c) Exercising means you're healthy.

7. Create a hypothesis for the following question: Does music have an affect on a person's mood?

8. Something that you keep the same in an experiment is called a(n)

- a) Variable
- b) Procedure
- c) Control/Constant

9. Sara wants to see if a new brand of hair dye lasts longer than the brand she currently uses. She puts the new hair dye on the left side of her head and the old brand on her right side. After 2 weeks she observes which side of her head has more gray hair showing through.

What is the independent variable?

- a) New hair dye
- b) Old hair dye
- c) Time the dye lasts

10. Sara wants to see if a new brand of hair dye lasts longer than the brand she currently uses. She puts the new hair dye on the left side of her head and the old brand on her right side. After 2 weeks she observes which side of her head has more gray hair showing through.

What is the dependent variable?

- a) New hair dye
- b) Old hair dye
- c) Time the dye lasts

11. Patrick loves bubble gum and would like to be able to blow bigger bubbles than anyone else in Bikini Bottom. To prepare for the Bikini Bottom Big Bubble Contest, he bought five different brands of bubble gum to see which one would have the biggest bubble.

Identify the variables in his experiment.

How many times he chews the gum ☐

☐ Independent Variable

The different brands of gum ☐

☐ Constant

Size of the bubble ☐

☐ Dependent Variable

12. Match each variable to the correct description based on the following scenario:

Kyle is testing how much sunlight affects plant growth. He has three different plants in different areas of his yard. He is giving each plant 5 mL of water every day.

Amount of Sunlight ☐

☐ Independent Variable

Plant Growth ☐

☐ Dependent Variable

Amount of Water ☐

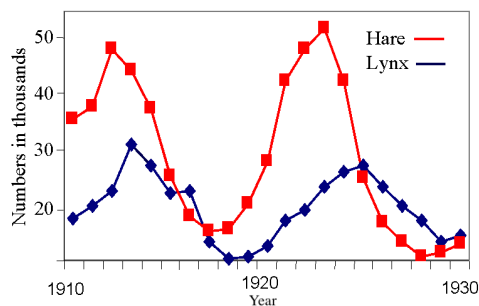
☐ Constant

13. SpongeBob loves to garden and wants to grow lots of pink flowers for his pal Sandy. He bought a special Flower Power fertilizer to see if will help plants produce more flowers. He plants two plants of the same size in separate containers with the same amount of potting soil. He places one plant in a sunny window and waters it every day with fertilized water. He places the other plant on a shelf in a closet and waters it with plain water every other day.

What did SpongeBob do wrong in this experiment?

14. Explain why it is important to only test one variable at a time? (Hint: Think of why we only have one independent variable at a time.)

15.



What is wrong with the following graph?

- a) X-axis scale is incorrect
- b) Missing a title
- c) Y-axis scale is incorrect
- d) None of the above

16.



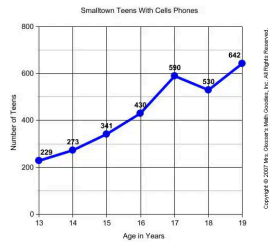
Identify the independent variable using the following graph

- a) The Park
- b) The Year
- c) Number of Visitors

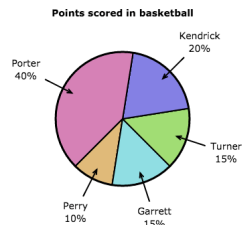
17. The Y axis is labeled with the

- a) Independent Variable
- b) Dependent Variable
- c) Constant Variable

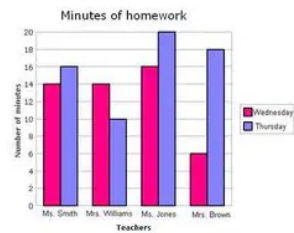
18. Match the graph to the correct description



○ show trends over time

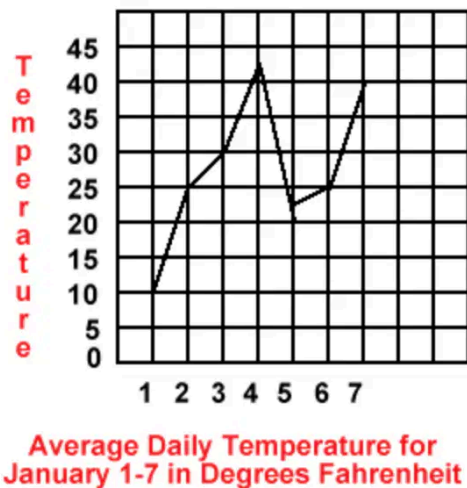


○ show a percentage of a whole



○ show a comparison between multiple groups

19.

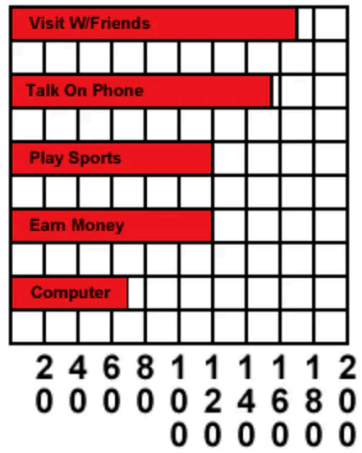


This graph shows the temperatures during the period of a

- a) Day
- b) Week
- c) Month

20.

**Favorite
Student
After
School
Activity**



Based on this survey would you say more students like to be around people or be alone after school?

a) Be alone

b) Be around people

21.

Name	Seashells collected
Sara	1
Zainab	3
Hoda	4
Mariam	2

Extra Credit:

The table shows the number of sea shells collected by some children.

If 100 shells were collected, predict the person that would collect the largest number of shells.

a) Sara

b) Zainab

c) Hoda

d) Mariam